AWS Website Deployment on Amazon EC2

Deploying a Secure Static Website Using Ubuntu Linux, Nginx & Route 53

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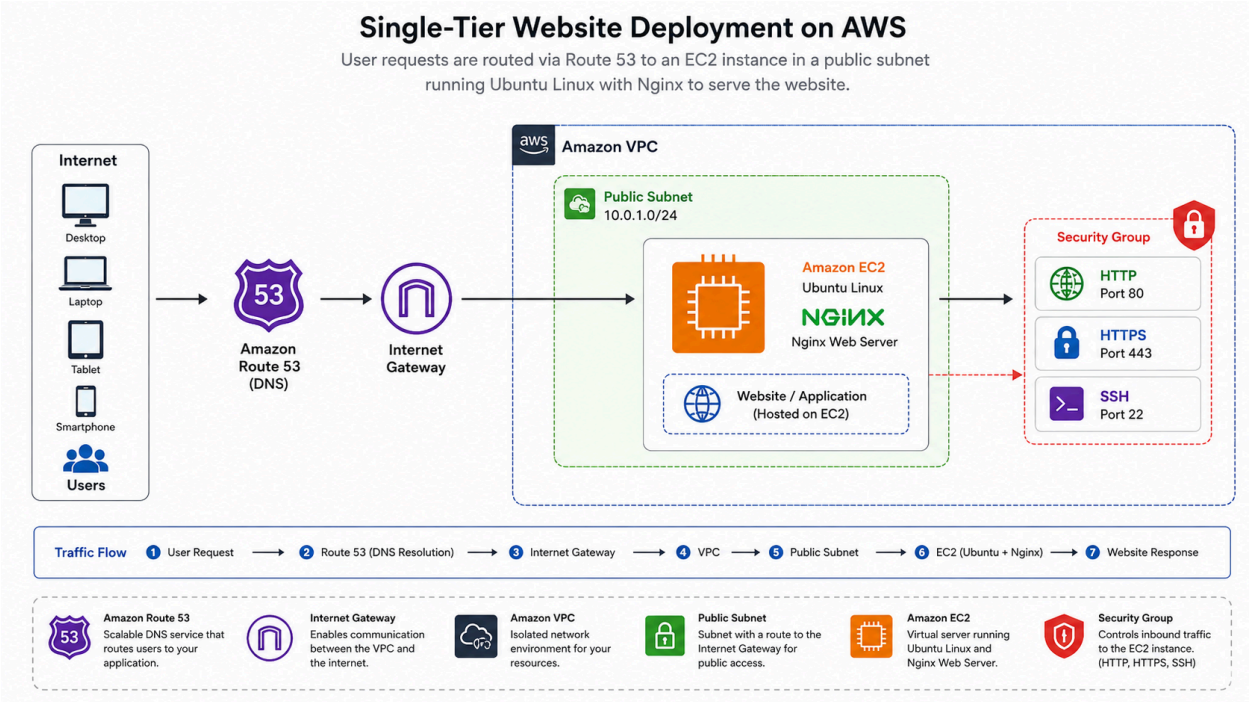
Project Overview

This project demonstrates the successful deployment of a secure static website on Amazon Web Services (AWS). The website was hosted on an Amazon EC2 instance running Ubuntu Linux with Nginx configured as the web server. Amazon Route 53 was used for DNS configuration, while AWS Security Groups were implemented to secure network access.

The project showcases practical experience in cloud infrastructure deployment, Linux server administration, web server configuration, DNS management, and AWS networking.



Technologies Used

Cloud Platform

- Amazon Web Services (AWS)

AWS Services

- Amazon EC2
- Amazon VPC
- Amazon Route 53
- AWS Security Groups

Operating System

- Ubuntu Linux

Web Server

- Nginx

Tools

- SSH
- Linux Terminal

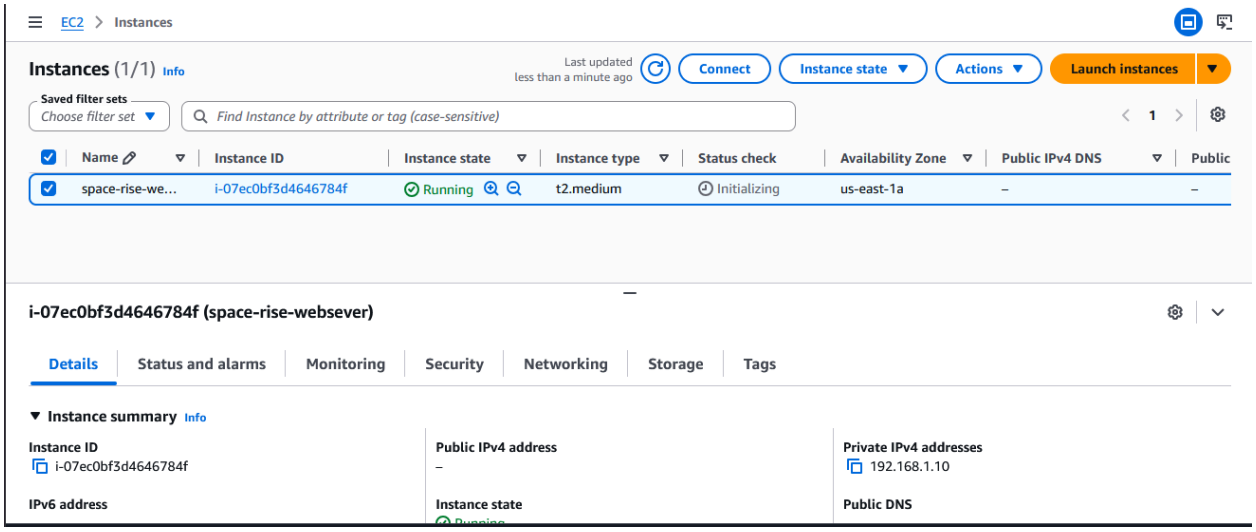
Project Objectives

- Deploy a static website on Amazon EC2.
- Configure Ubuntu Linux for web hosting.
- Install and configure Nginx as the web server.
- Configure DNS using Amazon Route 53.
- Secure the server using AWS Security Groups.
- Verify that the website is publicly accessible.

Implementation Process

Step 1 – Launch an Amazon EC2 Instance

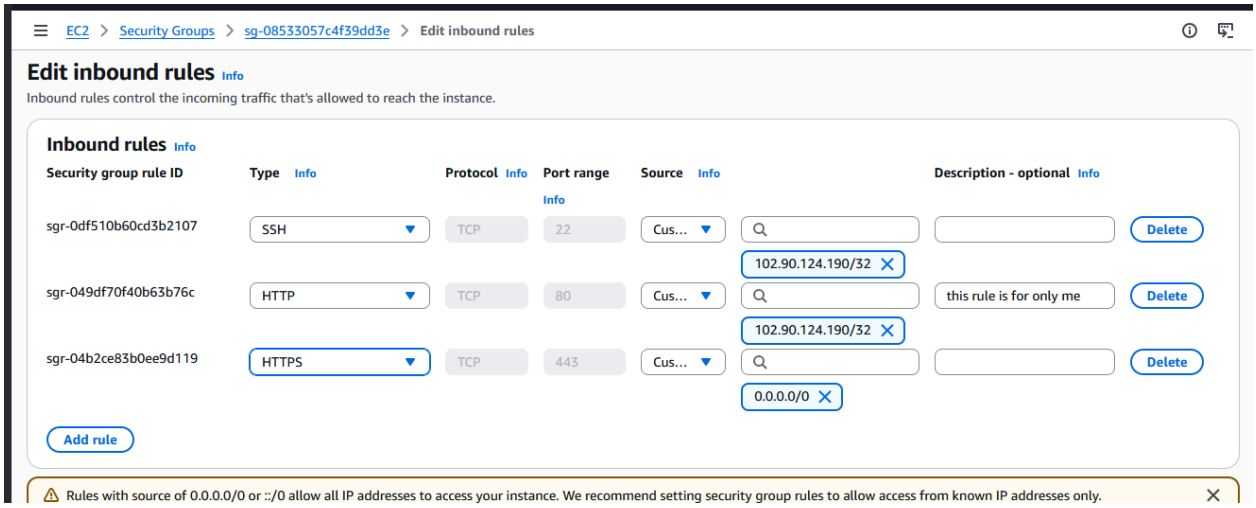
Created and configured an Ubuntu EC2 instance to host the website.



Step 2 – Configure Security Groups

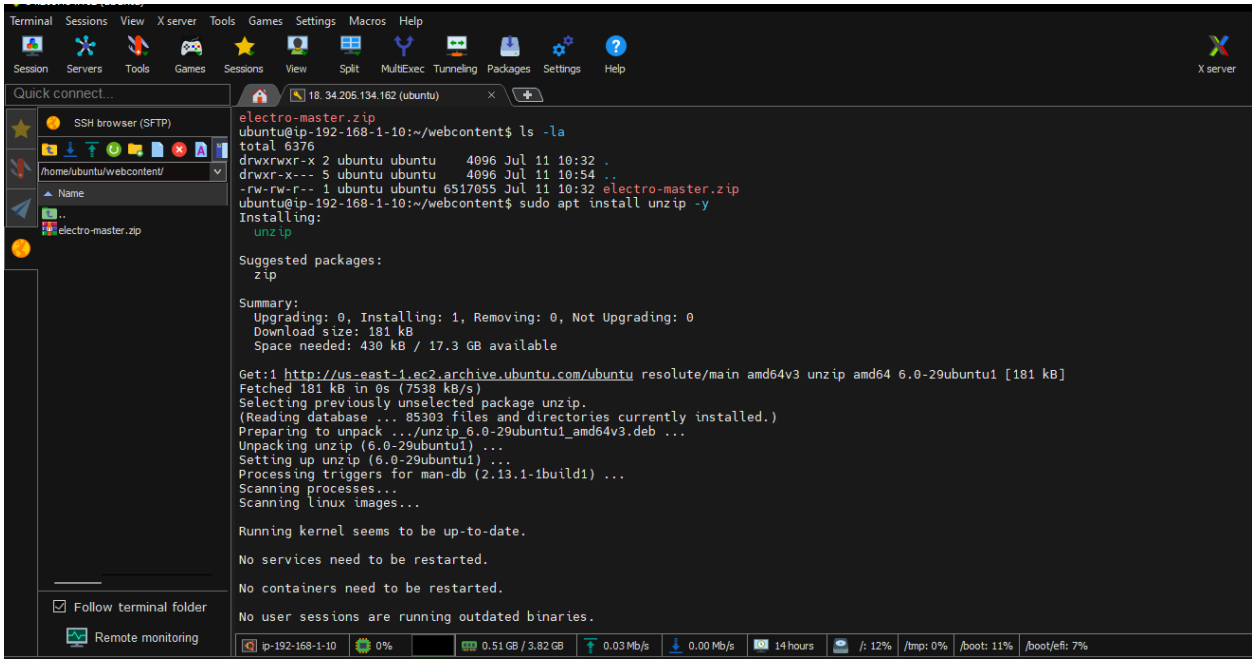
Configured inbound rules to allow:

- SSH (Port 22)
- HTTP (Port 80)
- HTTPS (Port 443)



Step 3 – Connect to the Server

Connected securely to the EC2 instance using SSH for remote administration.



Step 4 – Configure Ubuntu Linux

Updated the server packages and prepared the Linux environment for deployment.

Commands used:

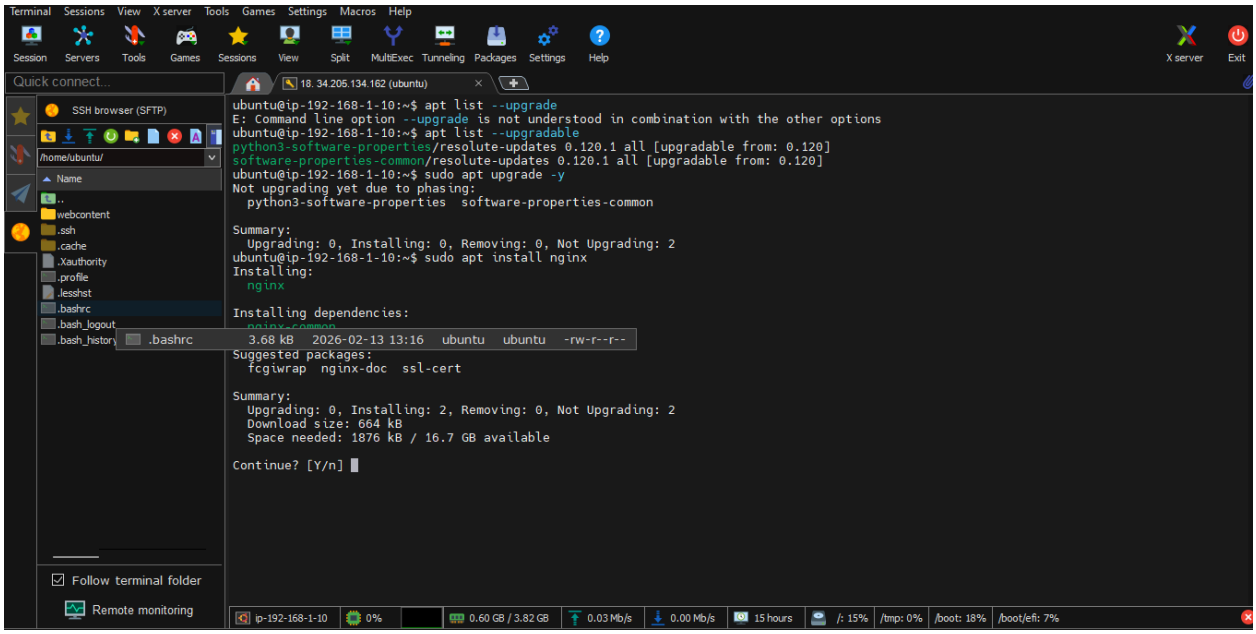
sudo apt update
sudo apt upgrade

Step 5 – Install Nginx

Installed and configured Nginx to serve the website.

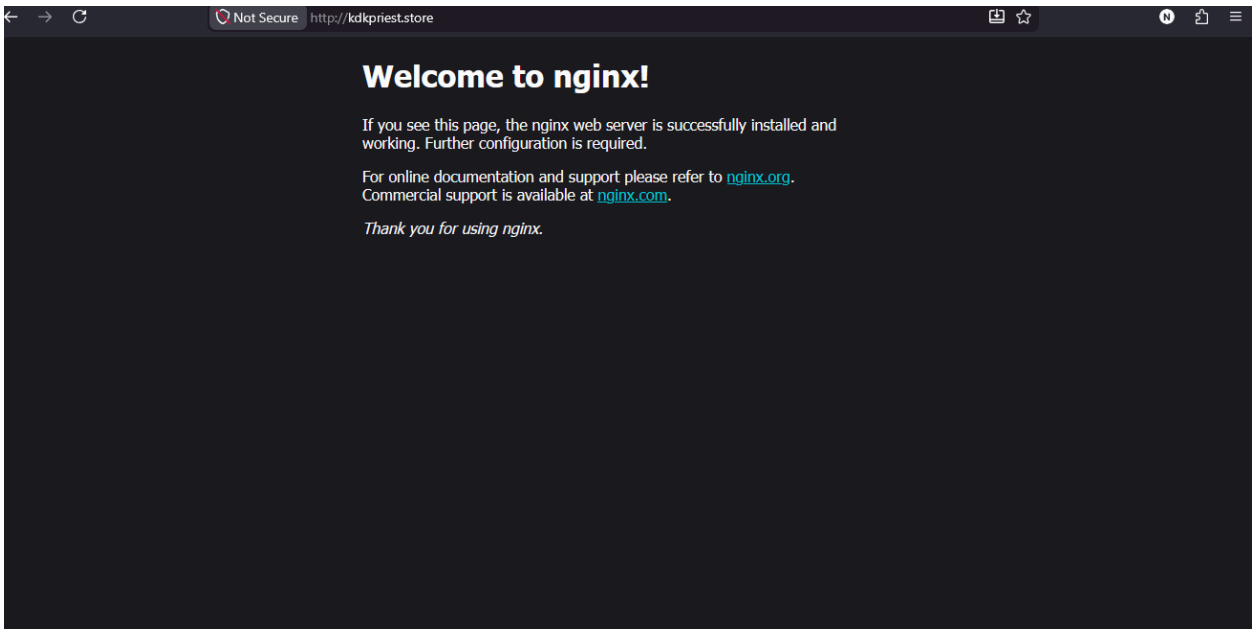
Commands used:

sudo apt install nginx
sudo systemctl start nginx
sudo systemctl enable nginx



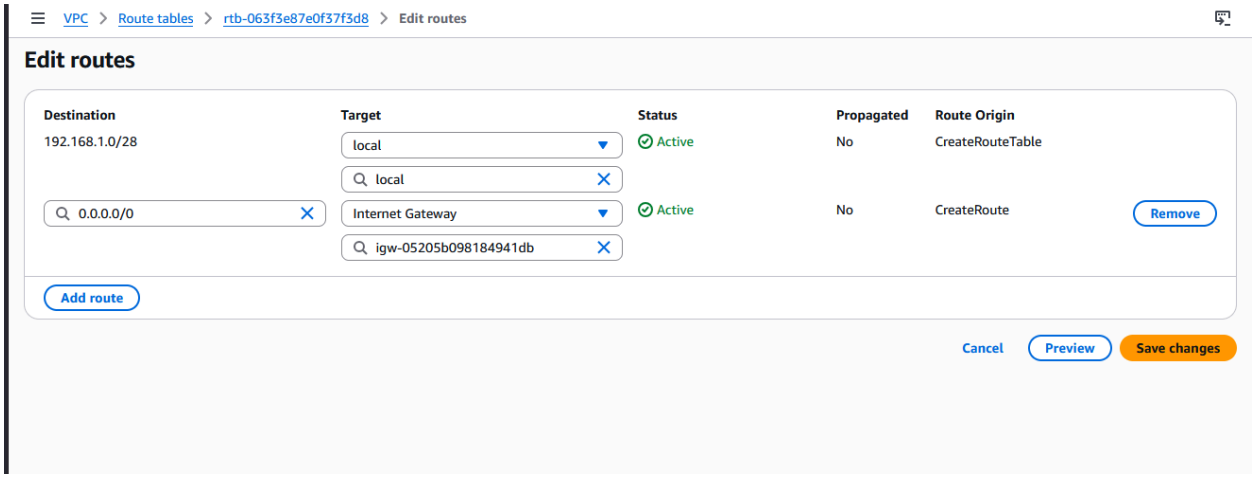
Step 6 – Deploy the Website

Uploaded the website files to the Nginx web root directory and verified successful deployment.



Step 7 – Configure Route 53

Configured DNS records to connect the domain name to the EC2 instance.



Project Outcome

The static website was successfully deployed on Amazon EC2 using Ubuntu Linux and Nginx. DNS resolution was configured using Amazon Route 53, and the infrastructure was secured using AWS Security Groups. The deployment demonstrates foundational cloud engineering skills and best practices for hosting static websites on AWS.

Skills Demonstrated

- AWS Cloud Computing
- Amazon EC2
- Amazon VPC
- Amazon Route 53
- Ubuntu Linux
- Linux Server Administration
- Nginx
- SSH
- DNS Configuration
- Website Deployment
- Cloud Networking
- Cloud Security

Deliverables

✔ Amazon EC2 Instance Deployment

✔ Ubuntu Linux Server Configuration

✔ Nginx Installation and Configuration

✔ Static Website Deployment

✔ Route 53 DNS Configuration

✔ AWS Security Group Configuration

✔ SSH Remote Administration

✔ Functional Website Verification

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Contact

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